

Nonlinear Optimization II

553.762 — Spring 2026

Course syllabus adapted from the public course website

1 Course overview

This course considers algorithms for solving various nonlinear optimization problems and, in parallel, develops the supporting theory. The primary focus will be on constrained optimization problems. Topics for the course will include: necessary and sufficient optimality conditions for constrained optimization; projected-gradient and two-phase accelerated subspace methods for bound-constrained optimization; simplex and interior-point methods for linear programming; duality theory; and penalty, augmented Lagrangian, sequential quadratic programming, and interior-point methods for general nonlinear programming. In addition, we will consider the Alternating Direction Method of Multipliers (ADMM), which is applicable to a huge range of problems including sparse inverse covariance estimation, consensus, and compressed sensing. Recommended Course Background: Multivariable Calculus, Linear Algebra, Real Analysis such as AS.110.405.

2 Meetings and course staff

Class meetings	Monday and Wednesday, noon–1:15 p.m.
Instructor office hours	Thursday, 3:00–4:30 p.m., Wyman S429
Teaching assistant	Pedro Izquierdo Lehmann
TA office hours	Tuesday, 9:15–10:00 a.m., Wyman S425
Teaching assistant	Thabo Samakhoana
TA office hours	Friday, 3:30–5:00 p.m., Wyman S425

3 Lecture sequence

The course website links the individual lecture notes. The order below follows that page, while the headings provide convenient sections for prototype assessments.

1. Introduction
2. Linear Programming
3. Nonlinear Programming Models
4. Fenchel and Lagrange Duality
5. Primal-Dual First Order Methods
6. Primal-Dual Second Order Methods

4 Homework and assessments

The course page lists about five homework assignments, with five linked at the time of this adaptation. Most assignments include at least one programming and testing task. Python is the preferred language and is used in demonstrations. Assignment materials and dates are provided on the [course site](#) and Canvas.

Assessment also includes a take-home midterm and take-home final. The course page directs students to Canvas for exam dates; the final date follows the official Johns Hopkins University exam schedule.

5 Course grade

The published policy uses four components: homework, midterm, final, and participation. Let their scores out of 100 be C_H , C_M , C_F , and C_P . For each student, the course selects the permissible weighting that gives that student the highest score. Homework, midterm, and final weights are H , M , and F ; the participation weight is $P = 100 - H - M - F$. The optimization stated on the course site is

$$\begin{aligned} & \max_{H, M, F \in \mathbb{R}} && C_H H + C_M M + C_F F + C_P(100 - H - M - F) \\ \text{subject to} &&& H + M + F \leq 100, \\ &&& H \geq 15, \quad M \geq 15, \\ &&& M \leq F, \\ &&& 50 \leq M + F \leq 80, \\ &&& H + M + F \geq 90. \end{aligned}$$

The weights are percentages. These constraints imply that participation receives between 0% and 10%. Students can earn full course marks without participation points. Participation credit may come from engagement in class, office hours, or substantive questions.

6 Reading

No single textbook is required; lecture notes appear on the course website. The page suggests the following optional references:

- J. Borwein and A. S. Lewis, *Convex Analysis and Nonlinear Optimization* (2006).
- J. Nocedal and S. Wright, *Numerical Optimization*, 2nd ed. (2006).
- R. T. Rockafellar, *Convex Analysis* (1970).
- J.-F. Bonnans and A. Shapiro, *Perturbation Analysis of Optimization Problems* (2000).
- R. T. Rockafellar and R. J.-B. Wets, *Variational Analysis* (1998).
- H. H. Bauschke and P. L. Combettes, *Convex Analysis and Monotone Operator Theory in Hilbert Spaces* (2011).
- D. Drusvyatskiy, *Convex Analysis and Nonsmooth Optimization*, (2020).